

Day 34

32-bit Up Counter in Verilog

Verilog Code

verilog code:

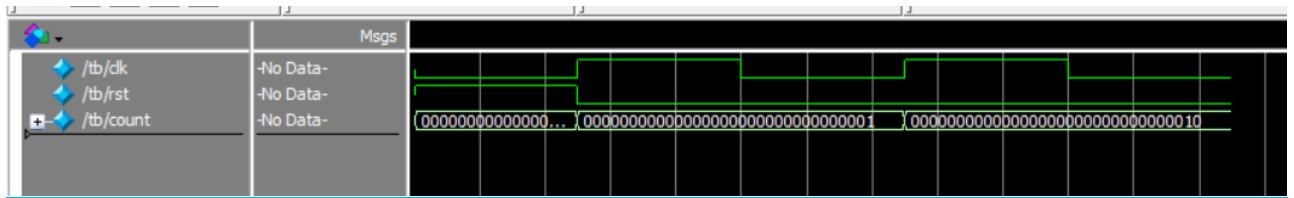
```
1  module up_counter(clk,rst,count);
2  input clk,rst;
3  output reg [31:0]count;
4  always @(posedge clk or posedge rst)
5  begin
6  if(rst)
7  count<=32'b0;
8  else
9  count<=count+1'b1;
10 end
11 endmodule
```

Testbench Code

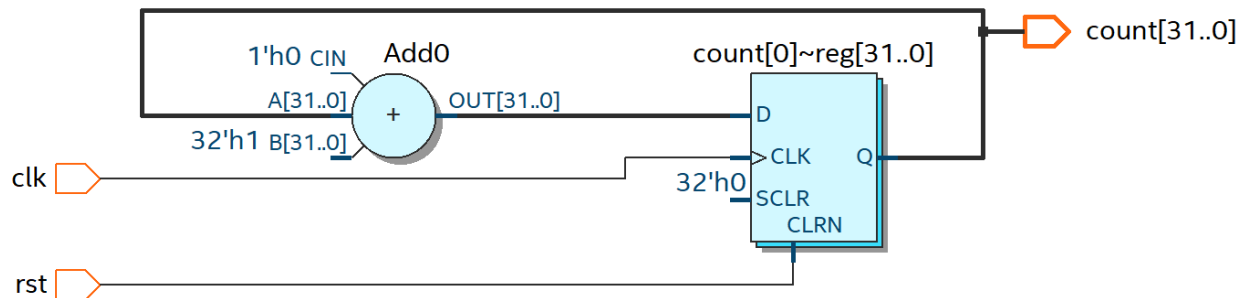
Tb code :

```
1  module tb;
2  reg clk,rst;
3  wire [31:0]count;
4  up_counter dut(clk,rst,count);
5  initial begin
6  clk=0;
7  forever #5 clk=~clk;
8  end
9  initial begin
10 rst=1; #5
11 rst=0; #20
12 $finish;
13 end
14 endmodule
15
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim